Vesicular transport across the cell membrane

Dr. Iman Nabil

Associate professor of Histology& Cell Biology

Alexandria Faculty of Medicine

Vesicular transport across the cell membrane

1- Definition of vesicular transport across the cell membrane.

2- Types of vesicular transport:

a) Endocytosis: Definition & subtypes.

b) Exocytosis: Definition& subtypes.

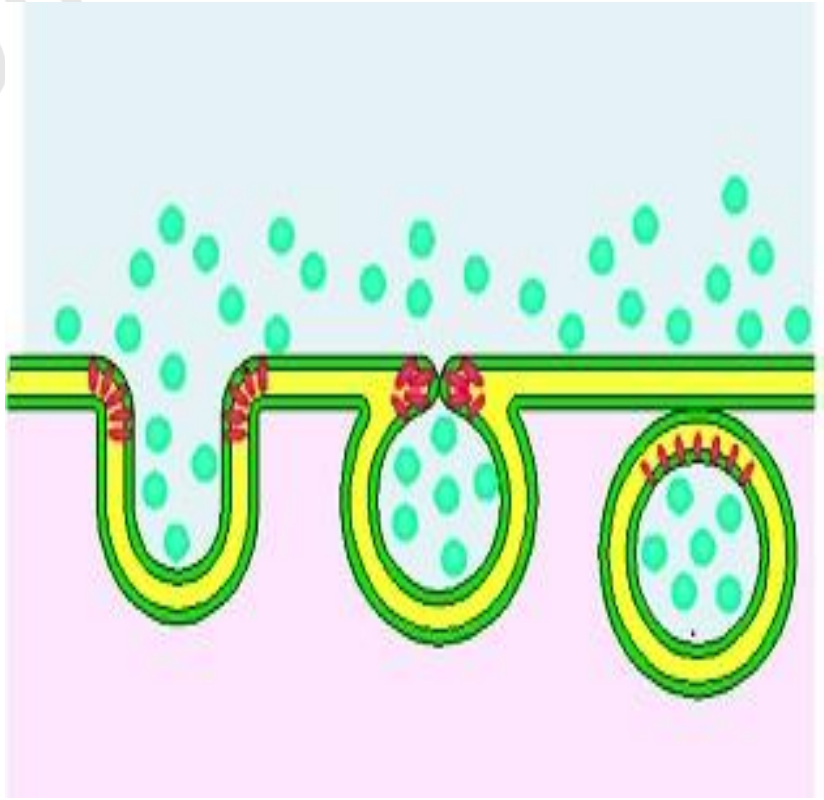
Dr Iman Nabil

Vesicular transport of macromolecules across the cell membrane

- Mass transfer of materials through the cell membrane occurs by formation of *vesicles*.
- It involves 2 processes:

1- Endocytosis.

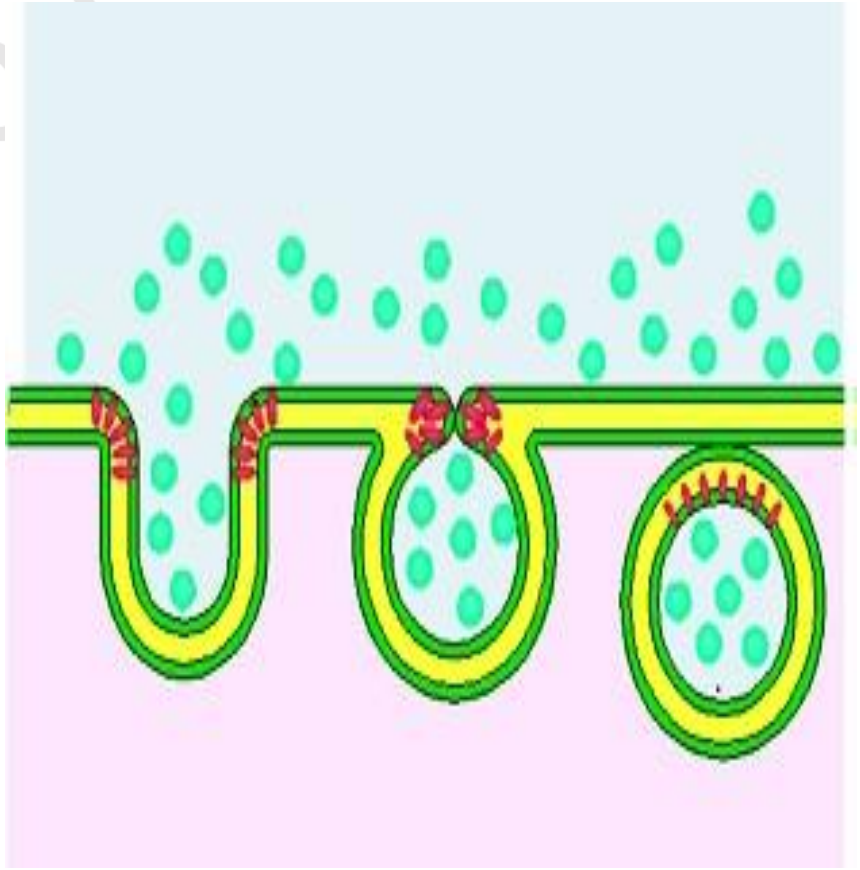
2- Exocytosis.



Dr Iman Nabil

1-Endocytosis

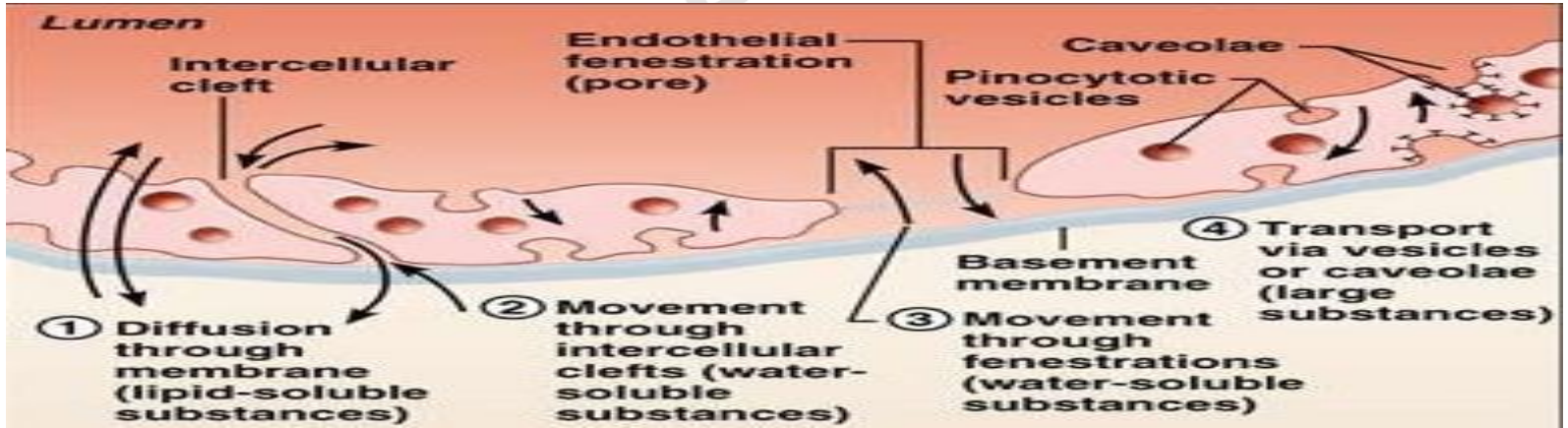
- Definition: the uptake of material from the extracellular space.
- It is an *active process* that involves invagination of the membrane to form a vesicle.
- 3 mechanisms of endocytosis are present in the cell

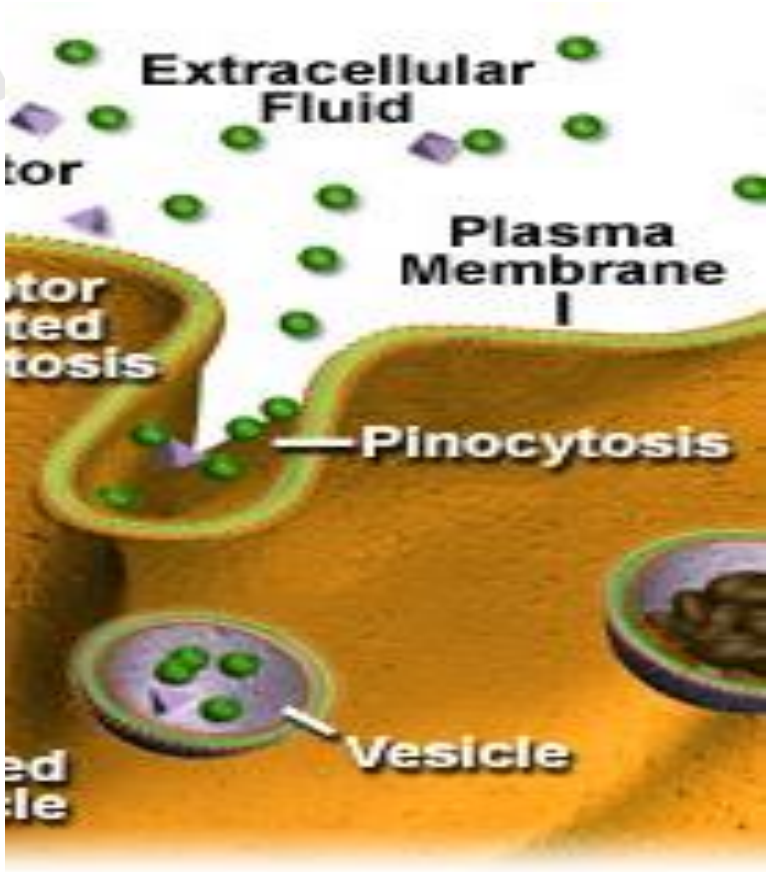


Dr Iman Nabil

A-Pinocytosis

- **Definition:** *a non-selective process*, occurs in nearly all cell types for uptake of fluid containing ions and small protein molecules.
- **Vesicle:** pinocytotic vesicles are small and have smooth surface.
- **Site:** most evident in the endothelium of blood vessels.





Dr Iman Nabil

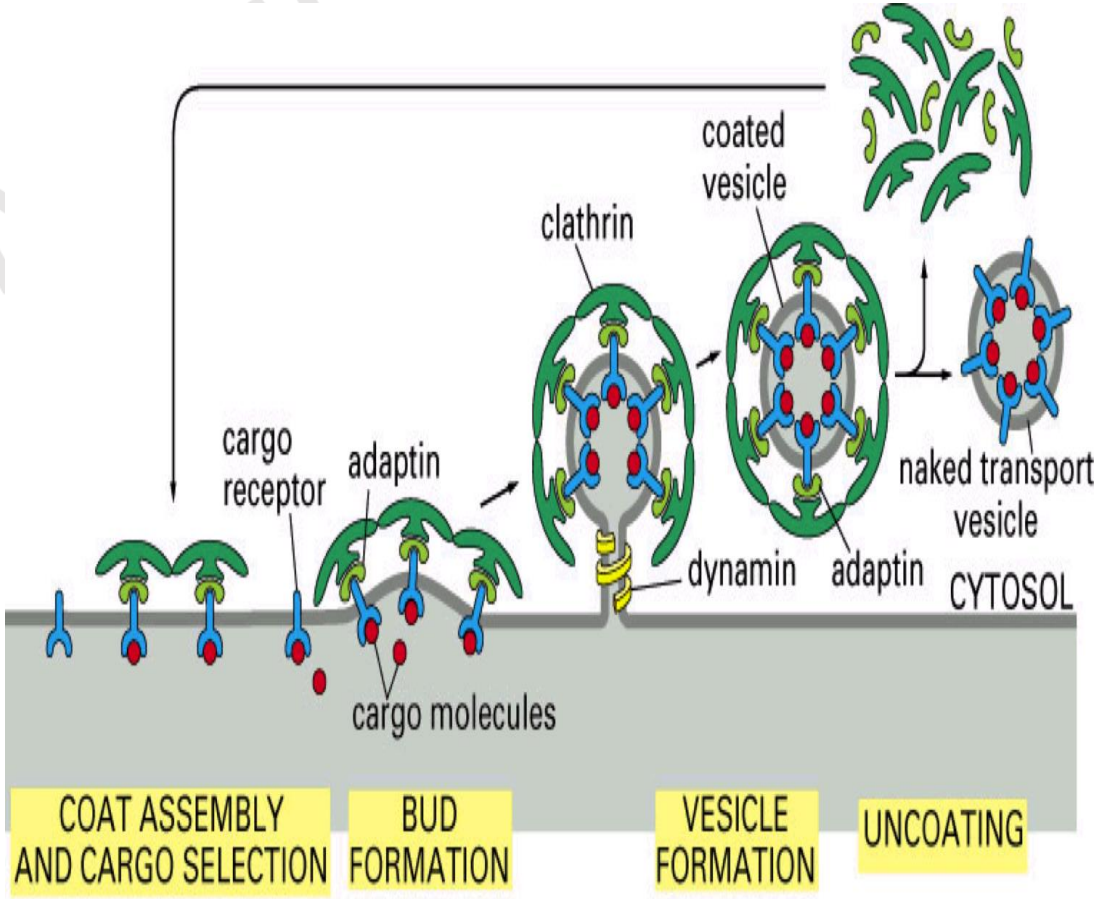
B-Receptor- mediated endocytosis

➢ **Definition**: *a highly-selective* process resulting in uptake of specific substances by a specific cell that has *receptors* for these substances e.g. protein hormones.

- These receptors are concentrated in specialized regions of the plasma membrane called *coated pits* (coated by clathrin).

1- When a substance binds to its receptor, clathrin-coated pits invaginate and give rise to *clathrin-coated vesicles* containing this specific substance.

2- Clathrin is lost and recycled leaving *uncoated vesicles*.



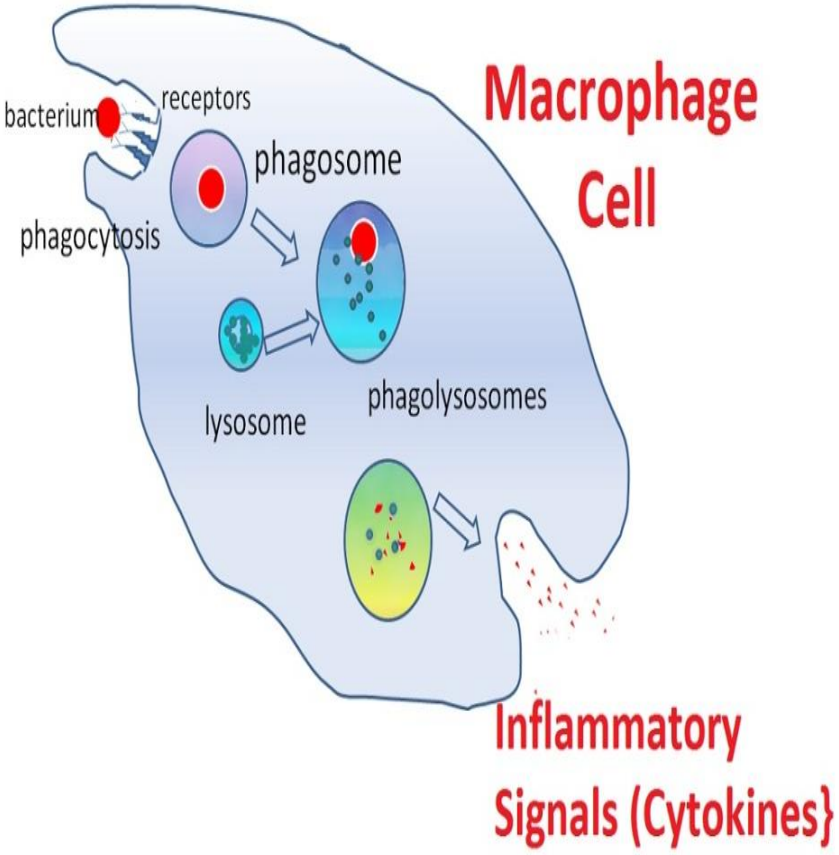
C-Phagocytosis

- Definition: ingestion of large solid particles, such as bacteria and cell debris, It is *a receptor-mediated* endocytosis *however*; it does not involve formation of coated pits or vesicles.
- Sites:phagocytes e.g. macrophages& neutrophils.

1- Binding of the receptor and foreign body results in extension of *pseudopodia* that engulf the particle.

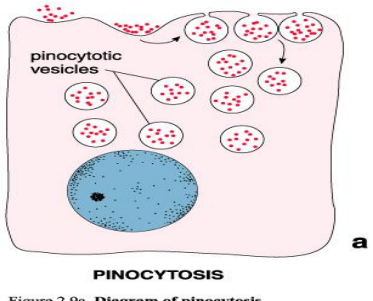
2- Fusion of the membrane to internalize the particle into the cytoplasm forming a *phagosome*.

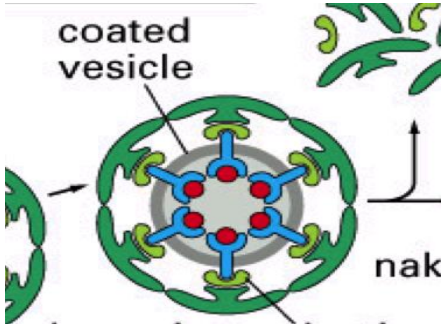
3- The contents of the phagosome are then digested through *lysosome*.

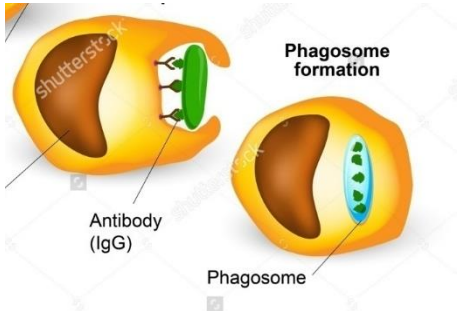


Dr Iman Nabi

Types of endocytosis	Pinocytosis	Receptor mediated endocytosis	Phagocytosis
1- Endocytosed material	Fluid containing ions & small molecules.	Specific substances (ligand) e.g. hormone.	Large solid particles e.g. bacteria.
2- Receptors for endocytosed material	Non selective.	Present .	Present .
3- Shape of the vesicle	Small& smooth.	Coated with clathrin.	No coated vesicle but the membrane fused to form phagosomes.
4- Type of cells	Nearly all cell types especially endothelium of blood vessels.	Specific cell that has receptor for specific substance.	Phagocytic cells.



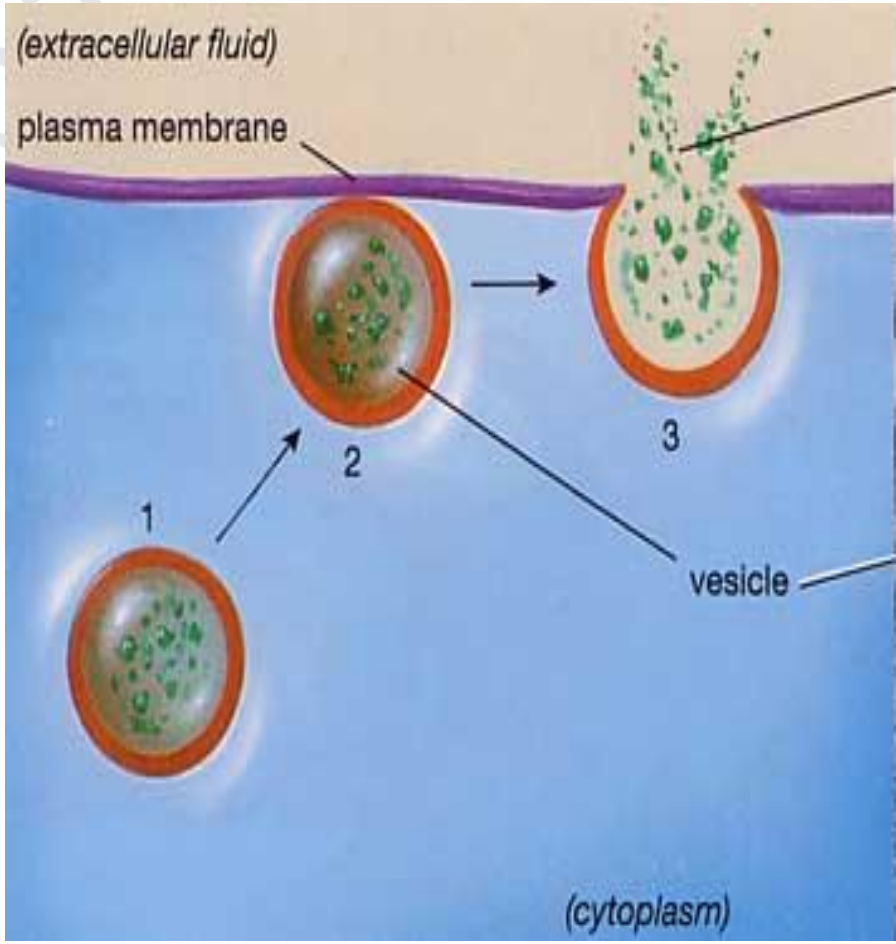




Dr Iman Nabil

2- Exocytosis

- **Definition**: the release of cell products into the extracellular space.
- During this process, a vesicle moves from the cytoplasm to the cell membrane, fuses with it and discharges its content.
- There are 2 types of exocytosis.



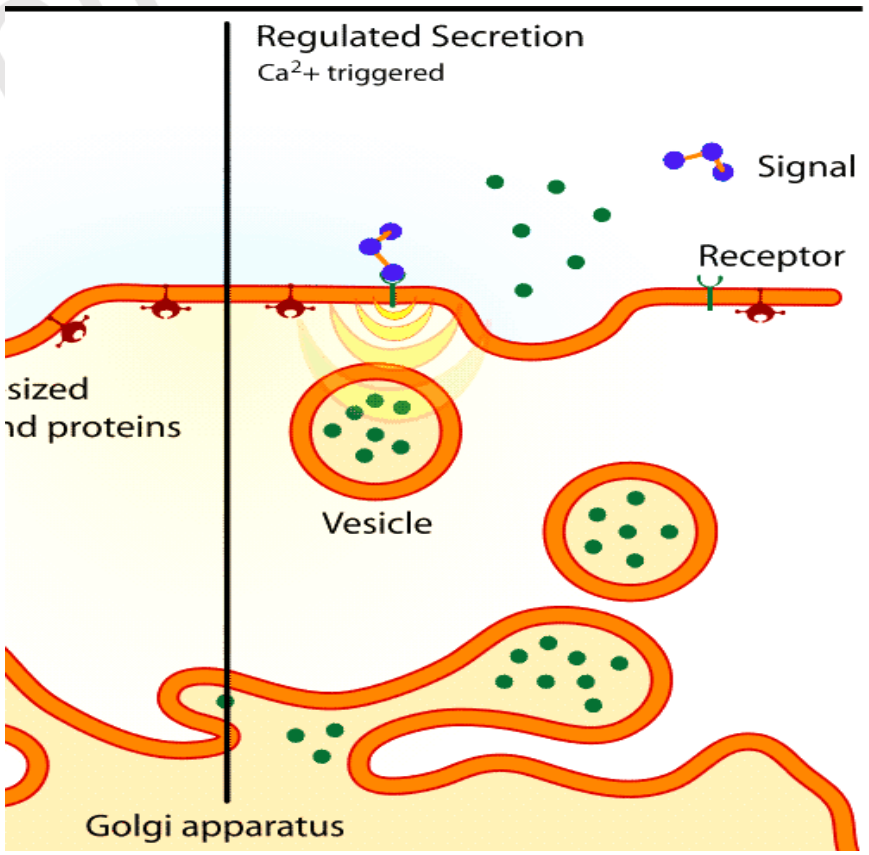
Dr Iman Nabil

A-Regulated secretion (stimulus- dependent)

1- The secretory products become *stored* forming *secretory granules*.

2- As a result of a *stimulus* (hormonal or neural stimulus), these vesicles move to the surface and fuse with the cell membrane to pour their contents outside the cell.

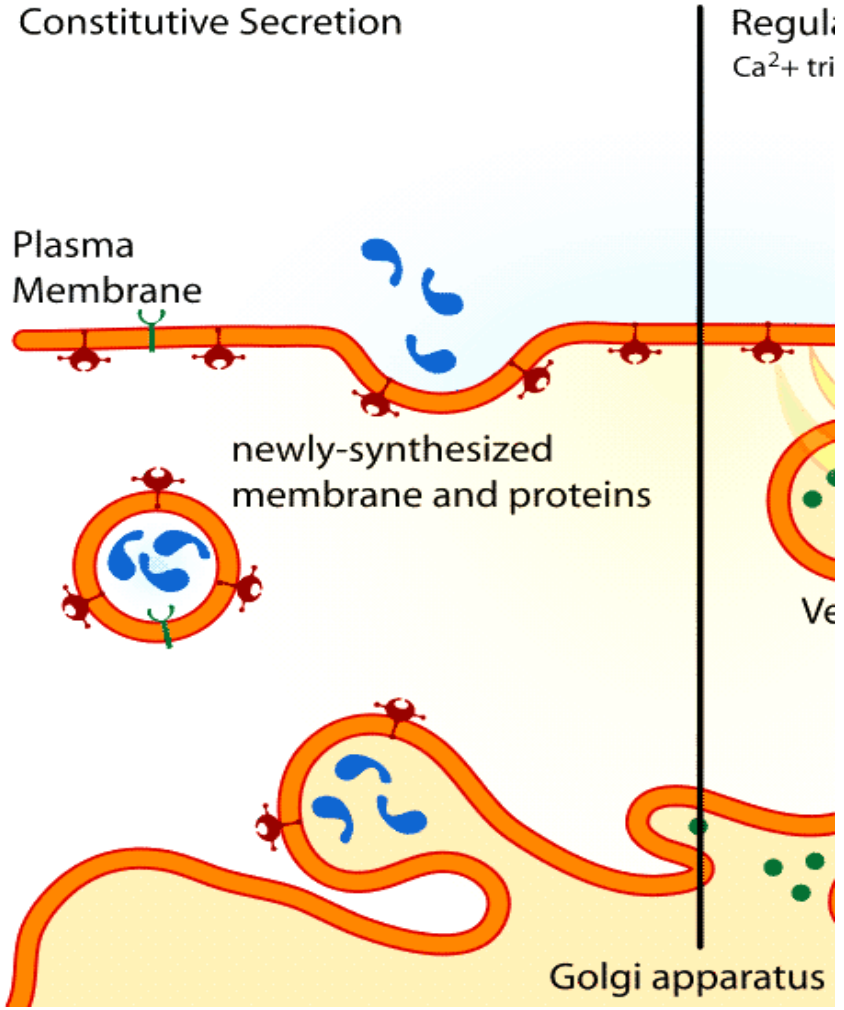
e.g. occurs during release of the digestive enzymes by the pancreas.



Dr Iman Nabil

B-Constitutive secretion

- The secretory products leave the cell *immediately* after their synthesis. These cells lack secretory granules.
- The secretion is released continuously through *secretory vesicles*.
- E.g. occurs during release of antibodies by plasma cells.



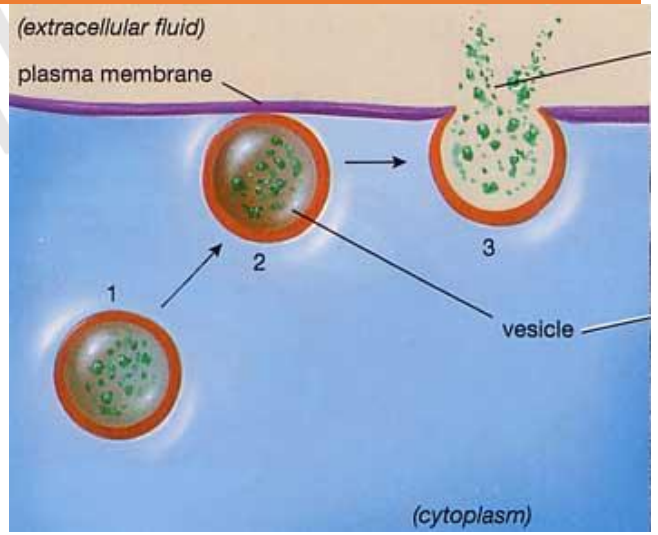
Dr Iman Nab

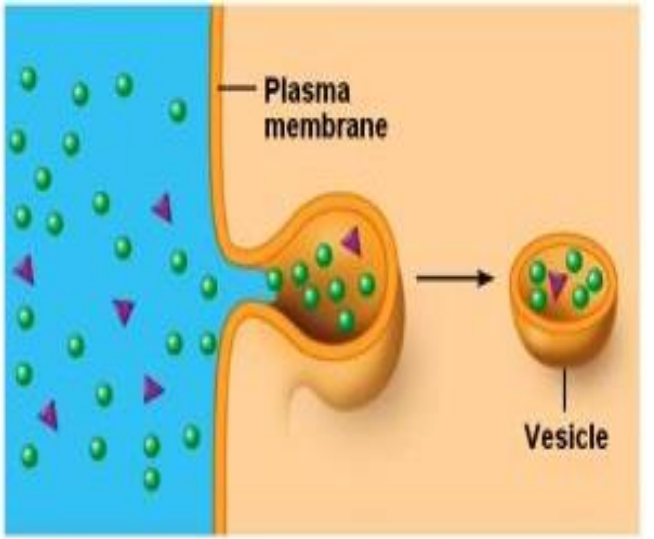
Types of exocytosis	Regulated secretion	Constitutive secretion
1- stimulus	Stimulus dependent.	No stimulus. They released continuously.
2- Secretory product	Concentrated & stored inside secretory granules	Leave the cell membrane immediately after their synthesis. No secretory granules.
3- Example of the released secretion	Digestive enzymes from pancreatic cell.	Antibodies from plasma cell, fibers secreted from fibroblast.

Dr Iman Nabil

Membrane recycling

- During the vesicular transport, the cell membrane is maintained; the excess membrane added to the cell membrane by *exocytosis* is constantly recycled again into the cytoplasmic compartments by *endocytosis*.





Dr Iman Nabil

Which of the following statement is the best to describe the receptor mediated endocytosis?

1- A non-selective process, occurs in nearly all cell types.

2- It involves binding of the receptor and the endocytosed material with subsequent extension of pseudopodia.

3- It occurs during the release of antibodies by plasma cells.

4-It give rise to clathrin-coated vesicles containing the specific substance.

Dr Iman Nabil

Which of the following statement is the best to describe the receptor mediated endocytosis?

1- A non-selective process, occurs in nearly all cell types.

2- It involves binding of the receptor and the endocytosed material with subsequent extension of pseudopodia.

3- It occurs during the release of antibodies by plasma cells.

4-It give rise to clathrin-coated vesicles containing the specific substance.

Dr Iman Nabil


THANK
YOU


Thank You